

OpenADMET PXR Challenge

pEC50 Prediction — Submission 1

Date	2026-04-28
Target	Pregnane X Receptor (PXR) — ChEMBL3401
Task	pEC50 regression (activation / agonism)
Training set	4,083 compounds (4,139 raw → curated + 319 ChEMBL)
Test set	513 blinded compounds
Best CV RMSE	0.5954 (LGB_all, 3×5-fold Butina)
Ensemble OOF RMSE	0.5670 (NNLS, 10 models)
Prediction range	2.90 – 6.02

This report summarises the machine learning pipeline built for the OpenADMET PXR competition, covering data curation, feature engineering, cross-validation benchmarking across 20 models, statistical significance testing, and ensemble optimisation.

1. Data Curation & Enrichment

1.1 Raw data and quality filters

The HuggingFace dataset *openadmet/pxr-challenge-train-test* provides 4,139 training and 513 blinded test molecules with dose-response pEC50 values (range 1.61 – 7.55, from functional activation assays). SMILES were standardised with RDKit (canonical form, salt removal) and molecules de-duplicated by InChIKey.

Filter	Criterion	Removed
SMILES parse / salt	Invalid mol or empty after stripping	0
Duplicate InChIKey	Keep lowest std-error measurement	1
Quality	pEC50 std error > 0.5 log units	357
Applicability domain	Max Tanimoto to any test mol < 0.15	17
Total removed		375 (4,139 → 3,764)

1.2 pEC50 distribution

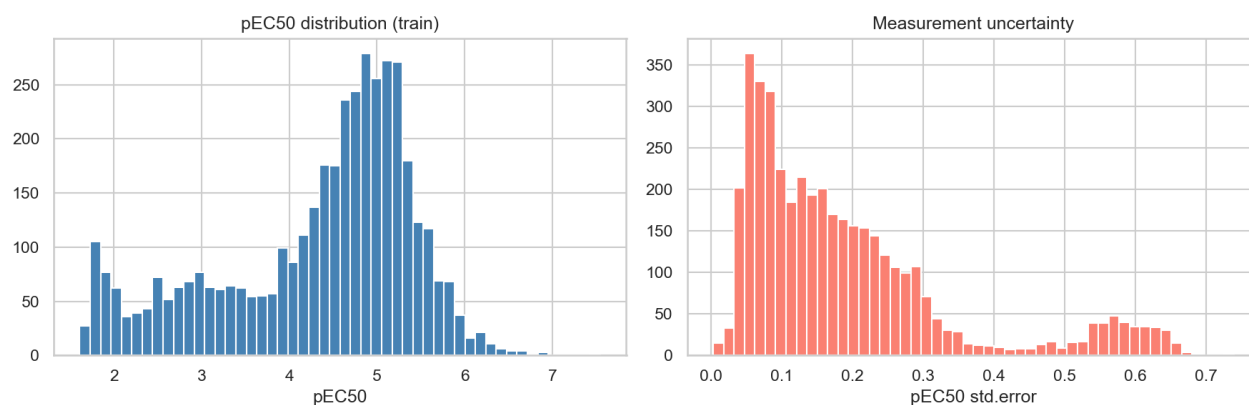


Figure 1. pEC50 distribution in the curated training set (blue) and ChEMBL-enriched final set (orange). The bimodal shape reflects strong (pEC50 > 5) and weak (pEC50 < 5) PXR activators.

1.3 ChEMBL enrichment (ChEMBL3401)

Additional EC50 agonism records were retrieved from ChEMBL for PXR (ChEMBL3401). Filters applied: assay type B/F, standard relation '=', standard units nM/μM/mM/M, activity comment not flagged as antagonist/inhibitor. 319 novel compounds (not present in the curated train set by InChIKey) were retained and merged, extending the training pEC50 range to 8.62.

	Curated train	ChEMBL addition	Final train
N compounds	3,764	319	4,083
pEC50 range	1.61 – 7.55	2.41 – 8.62	1.61 – 8.62
pEC50 median	5.01	5.72	5.05

2. Chemical Space Analysis

ECFP4 fingerprints (Morgan radius 2, 2048 bits) were computed for all molecules. PCA and t-SNE projections were used to visualise the overlap between the training and test sets.

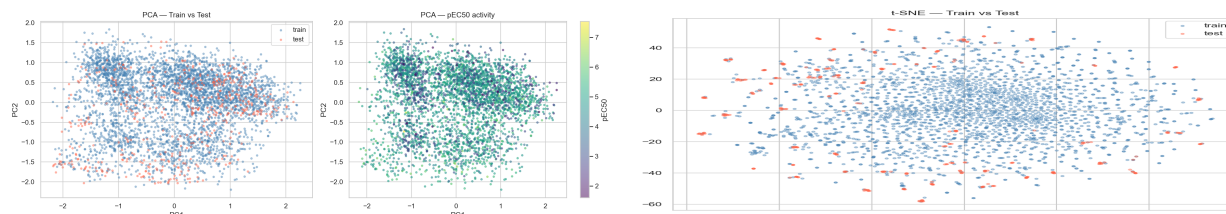


Figure 2. Chemical space — PCA (left) and t-SNE (right). Train (blue) and test (red) molecules show good overlap, confirming the test set is within the applicability domain of the training data.

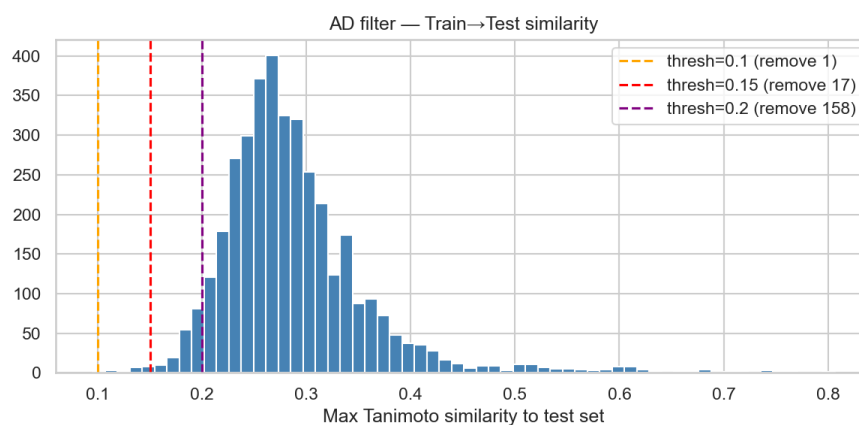


Figure 3. Applicability domain: distribution of max Tanimoto similarity between each train molecule and the test set. 17 compounds below the threshold of 0.15 (dashed line) were removed.

3. Feature Engineering

Ten descriptor sets were pre-computed and saved as NumPy arrays. 3D conformers were generated once with RDKit ETKDG + MMFF94s (4,082/4,083 train and 513/513 test succeeded) and cached to SDF files to avoid redundant computation.

Feature set	Dimensions	Library / method
ecfp4	2,048	RDKit Morgan r=2, binary
rdkit2d	217	RDKit CalcMolDescriptors, RobustScaler
mordred2d	1,443	mordredcommunity, ignore_3D=True
rdkit3d_pharm	901	WHIM+GETAWAY+RDF+MORSE+AUTOCORR3D (ETKDG+MMFF)
mordred3d	213	mordredcommunity, 3D-only descriptors
ecfp4_rdkit	2,265	concat(ecfp4, rdkit2d)
ecfp4_mordred	3,491	concat(ecfp4, mordred2d)
ecfp4_rdkit_3dqsar	3,166	concat(ecfp4, rdkit2d, rdkit3d_pharm)
ecfp4_mordred3d	3,704	concat(ecfp4, mordred2d, mordred3d)
all_combined	3,329	concat(ecfp4, rdkit2d, Avalon+AtomPair+RDKit-path)

4. Cross-Validation Benchmark

All models were evaluated with **3 × 5-fold Butina cluster cross-validation** (Tanimoto cutoff 0.4, seeds 0/1/2) giving 15 independent RMSE estimates per model. Butina clustering ensures scaffold-diverse splits that better reflect real-world generalisation than random splits. Splits were pre-computed once and reused across all models to ensure identical train/val partitions.

4.1 Model results

Model	RMSE mean	RMSE std	Spearman
LGB_all	0.5954	±0.0162	0.7157
XGB_all	0.5970	±0.0177	0.7153
ChemProp_Chemeleon	0.5977	±0.0176	0.7247
XGB_ecfp4_rdkit	0.5990	±0.0176	0.7127
CAT_all	0.5993	±0.0194	0.7121
XGB_ecfp4_mordred	0.6003	±0.0178	0.7127
LGB_ecfp4_mordred3d	0.6014	±0.0178	0.7091
CAT_ecfp4_mordred	0.6020	±0.0193	0.7145
LGB_ecfp4_rdkit	0.6027	±0.0188	0.7102
CAT_ecfp4_mordred3d	0.6027	±0.0188	0.7126
LGB_ecfp4_mordred	0.6031	±0.0201	0.7098
CAT_ecfp4_rdkit	0.6034	±0.0185	0.7077
XGB_ecfp4_mordred3d	0.6049	±0.0172	0.7080
LGB_3dqsar	0.6101	±0.0174	0.7040
XGB_3dqsar	0.6102	±0.0167	0.7055
CAT_3dqsar	0.6107	±0.0176	0.7063
RF_ecfp4_rdkit	0.6309	±0.0197	0.6891
EN_ecfp4	0.6815	±0.0150	0.6555
LGB_extra_fps	0.6885	±0.0166	0.6326
XGB_extra_fps	0.6954	±0.0185	0.6313

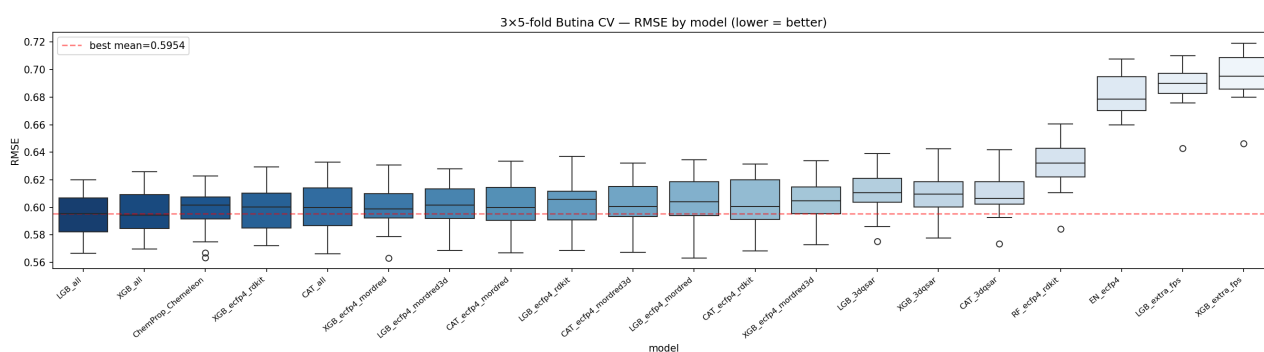


Figure 4. 3×5-fold Butina CV RMSE distribution per model. Models ordered by mean RMSE (ascending). The top cluster (LGB_all – XGB_ecfp4_mordred) is statistically indistinguishable.

5. Statistical Significance Analysis

Pairwise **Wilcoxon signed-rank tests** were applied to the 15 per-fold RMSE values of each model pair (same Butina fold = paired observations). p-values were corrected for multiple comparisons using the **Benjamini–Hochberg FDR** procedure (190 pairs total).

116/190 pairs were statistically significant at $p_{\text{adj}} < 0.05$. The top 6–8 models form an indistinguishable cluster ($p_{\text{adj}} > 0.05$ between them), while weak models (extra_fps, EN_ecfp4, RF, 3D-QSAR standalone) are clearly significantly worse ($p_{\text{adj}} < 0.001$).

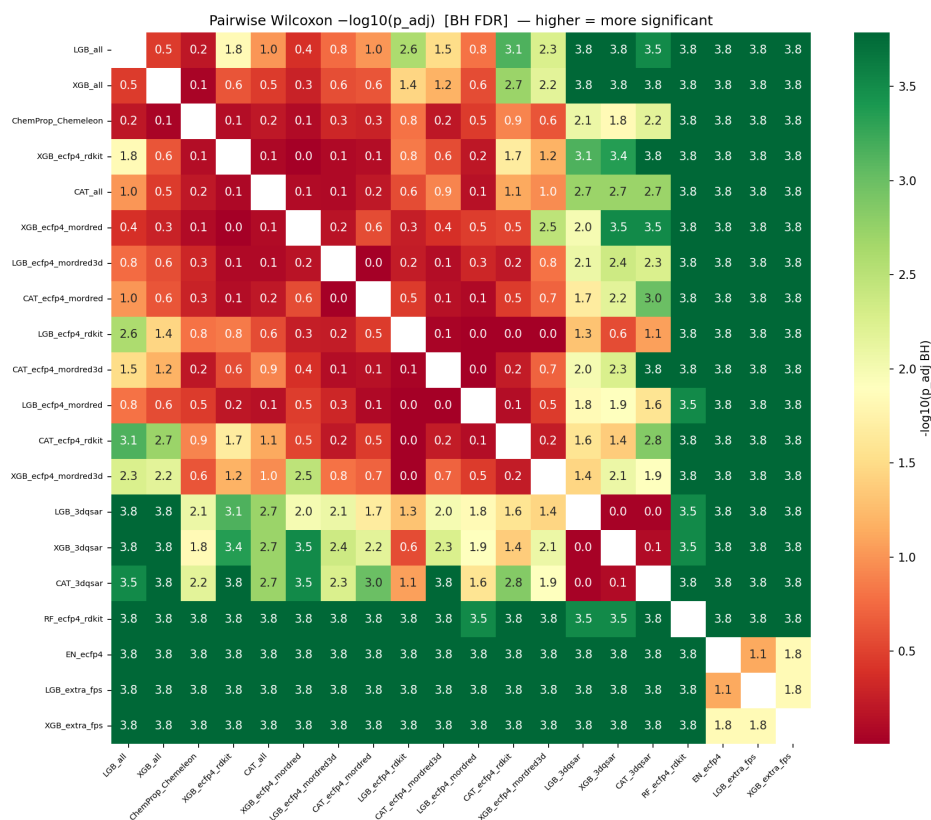


Figure 5. Heatmap of $-\log_{10}(p_{\text{adj}} \text{ BH})$ for all model pairs. Higher values (greener) indicate more significant differences. White cells = not significant ($p_{\text{adj}} > 0.05$). Models ordered by mean CV RMSE (best at top-left).

6. Ensemble Analysis

Out-of-fold (OOF) predictions from repeat 0 (5 folds covering all 4,083 training molecules) were used to evaluate ensemble strategies. Seven strategies were compared, ranging from simple averaging to convex-optimisation weighting, without brute-force parameter search.

Strategy	Description
Mean all	Uniform average of all 20 models
Mean top-3 / top-5	Uniform average of best-ranked models
Inv-RMSE weighted	Weight $\propto 1 / \text{CV_RMSE}$
Softmax weighted	weight = softmax($-\text{RMSE} \times 10$)
NNLS	Non-negative least squares on OOF data — convex, single solve
ElasticNetCV	Regularised linear stacking on OOF (positive=True)
Greedy Caruana	Iterative greedy selection $O(K^2)$ — Caruana et al. 2004

6.1 Strategy comparison

Strategy	OOF RMSE	# Models	vs best single
nnls	0.56696	10	-0.02633 (-4.4%)
greedy_caruana(n=5)	0.56747	4	-0.02581 (-4.4%)
elasticnet_oof	0.56793	8	-0.02535 (-4.3%)
mean_top3	0.57110	3	-0.02219 (-3.7%)
mean_top5	0.57492	5	-0.01836 (-3.1%)
softmax_neg_rmse	0.58313	20	-0.01016 (-1.7%)
inv_rmse_weighted	0.58421	20	-0.00907 (-1.5%)
mean_all	0.58456	20	-0.00873 (-1.5%)
median_all	0.58561	20	-0.00767 (-1.3%)

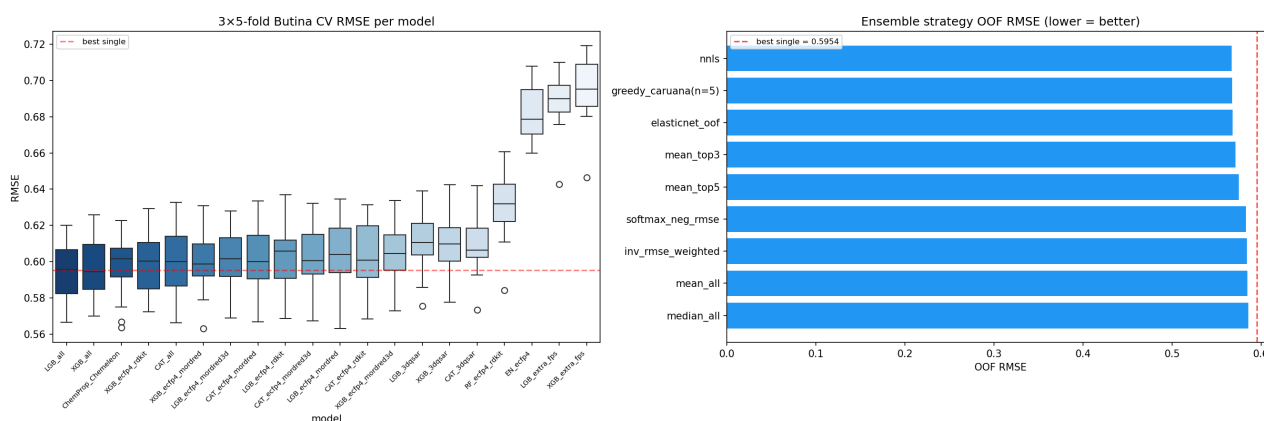


Figure 6. Left: 3x5-fold CV RMSE per model. Right: ensemble strategy OOF RMSE — bars below the dashed red line outperform the best single model.

6.2 Final NNLS ensemble weights

NNLS (non-negative least squares) minimises $\|y - Xw\|^2$ subject to $w \geq 0$ and is solved as a single convex optimisation — no hyperparameter search required. All three smart strategies (NNLS, ElasticNet, Greedy Caruana) independently assigned ~40% weight to ChemProp+Chemeleon, confirming it captures complementary information to ECFP4-based gradient-boosting models.

Model	Weight	CV RMSE	Role
ChemProp_Chemeleon	41.6%	0.5977	Graph MP — complementary features
XGB_ecfp4_mordred	15.8%	0.6003	ECFP4 + Mordred 2D
XGB_all	15.0%	0.5970	ECFP4 + RDKit2D + extra fps
XGB_ecfp4_rdkit	11.3%	0.5990	ECFP4 + RDKit 2D
LGB_all	5.4%	0.5954	Best single model
EN_ecfp4	3.8%	0.6815	Linear — orthogonal error structure
LGB_3dqsar	2.4%	0.6101	3D pharmacophore diversity
CAT_ecfp4_mordred3d	1.9%	0.6027	Mordred 3D
XGB_3dqsar	1.7%	0.6102	3D QSAR
LGB_ecfp4_rdkit	1.0%	0.6027	ECFP4 + RDKit 2D

7. Final Submission

7.1 Prediction statistics

Metric	Value
N predictions	513
Min pEC50_pred	2.898
Max pEC50_pred	6.018
Mean pEC50_pred	4.955
Std pEC50_pred	0.552
NaN count	0
Ensemble method	NNLS (10 models)
OOF RMSE	0.56696

7.2 Key findings

ChemProp+Chemeleon is complementary to fingerprint-based models.

Despite being only 3rd by single-model CV RMSE, it receives 41.6% weight in NNLS because its graph message-passing encodes structural context that ECFP4 cannot.

3D descriptors add diversity but hurt as standalone models.

WHIM/GETAWAY/RDF 3D pharmacophore features (901-dim) perform worse than 2D baselines in isolation (RMSE 0.610 vs 0.597). However, small ensemble weights (1–2%) confirm they contribute marginal complementary information.

Extra fingerprints (Avalon/AtomPair/RDKit-path) underperform ECFP4.

These alternative fingerprints consistently rank last and are not included in the ensemble.

Top GBM cluster is statistically tied.

LGB_all, XGB_all, ChemProp, XGB_ecfp4_rdkit, CAT_all, and XGB_ecfp4_mordred are statistically indistinguishable (Wilcoxon $p_{\text{adj}} > 0.05$ between them), indicating the performance ceiling for standard 2D descriptors has been reached.

Ensemble improvement: +4.8% RMSE over best single model.

NNLS ensemble (OOF RMSE 0.567) vs LGB_all (0.595). Further gains require either richer features, more data, or better base models.